Model B Fit to the Real Data

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1 Summary

Model B (hierarchical over sessions, with the design-based priors described in the convergence report) was fit to all 12,937 trials of the Sirius data (42 sessions, 4 groups). The main results:

- **The distractor lowers the PSE.** In the bimanual condition the monkey reports "high" at lower amplitudes. Averaged over the 42 sessions the PSE shift (bimanual − unimanual) is -4.3 μm [-5.0, -3.5] for the left hand and -6.1 μm [-6.8, -5.3] for the right hand. Its posterior mean is negative in 40 of 42 sessions for the left hand and in all 42 for the right hand, and its 95% HDI excludes 0 in 23 (left) and 32 (right) sessions.
- **The distractor raises** γ_h, the rate of "high" responses to the weakest stimuli, by about 0.02 (left) and 0.035 (right). For the right hand the HDI excludes 0 in all 42 sessions.
- **The distractor slightly raises the JND** (less precise discrimination), clearly for the left hand ($+1.0$ μm [0.5, 1.6], largely from a few sessions) and weakly for the right hand ($+0.5$ μm [-0.1, 1.0]). It has no effect on γ_l.
- **The PSE shift grows with distractor amplitude** (exploratory, Section 4.1): about -2 μm per 10 μm of distractor amplitude for both hands.
- The sampler converged, the model reproduces the data well, and the new priors change the hyperparameter estimates only slightly from the earlier fit.

Every figure and number in this report is produced by `Main_Analysis_B.py`; the cell "Numbers for Fit_B_Report.tex" prints the numbers that the earlier cells do not. All intervals are 95% highest density intervals (HDIs). PSE and JND are in micrometres of stimulus amplitude: $\text{PSE}_{\mu m} = 28 + 14.63\,\text{PSE}$ and $\text{JND}_{\mu m} = 14.63\,\text{JND}$, where 28 and 14.63 are the mean and sd of the 8 amplitude levels.

2 Model and Fit

The model is the one described in the Model B convergence report: per-session psychometric parameters $(\beta_0, \beta_1, \gamma_h, \gamma_l)$ for each group, drawn from group-level distributions whose means and sds have the design-based priors $\mu_{\beta_0} \sim \mathcal{N}(0, 2)$, $\sigma_{\beta_0} \sim \text{HalfNormal}(2)$, $\mu_{\beta_1} \sim \text{LogNormal}(\log 4, 0.5)$, $\sigma_{\beta_1} \sim \text{Exponential}(1)$, $\mu_\gamma \sim \mathcal{N}(-3, 1.5)$ and $\sigma_\gamma \sim \text{HalfNormal}(0.5)$.

The fit (`Main_Analysis_B.py`) used NUTS with 4 chains of 1000 tuning and 5000 kept draws (20,000 draws in total) and took 31 minutes.

2.1 Sampling diagnostics

There were no divergences, and the tree depth never exceeded 6. Of the 2,080 parameters, all but four have $\hat{R} = 1.00$, bulk ESS $\geq$ 2,210 and tail ESS $\geq$ 4,527.

The four exceptions are the raw variables `z_sig_b0` ($\hat{R}$ up to 1.73), and they are harmless. The model sets $\sigma_{\beta_0} = |2\,z|$ with $z \sim \mathcal{N}(0, 1)$, so z and $-z$ give the same model. The data keep σ_{β_0} well away from 0, so each chain stays on the sign it started on, and the chains disagree only about the sign. σ_{β_0} itself has $\hat{R} = 1.00$ and bulk ESS $\geq$ 7,000. Defining σ_{β_0} directly as a `HalfNormal(2)` variable (as is effectively already done for σ_{β_1}) would remove the warning without changing the model.

Figures 1 and 2 show trace plots for a few of the parameters that `Main_Analysis_B.py` singles out: the four chains overlap completely and mix quickly. This includes the lapse rates of left_bi on 05-30, which have the longest right tails of any session, and the lapse hyperparameters of left_bi, whose sd σ_{γ_h} piles up against 0.

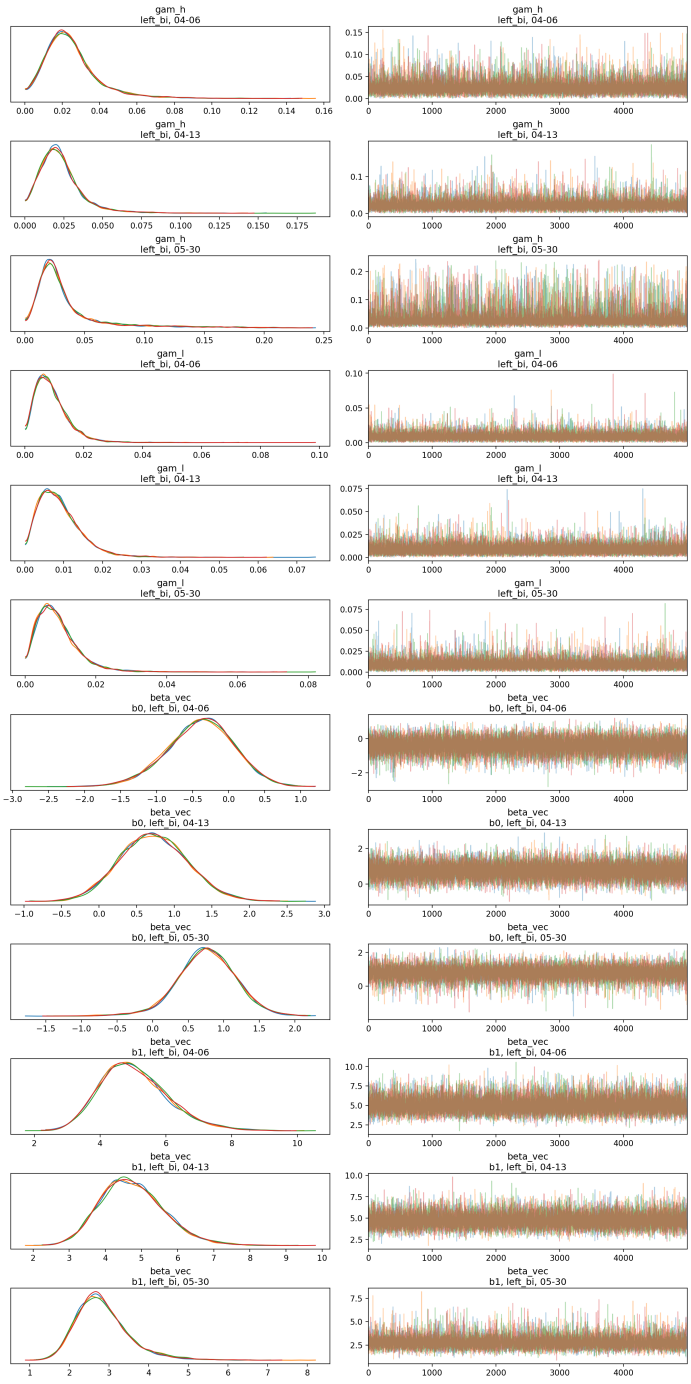


Figure 1: Trace plots (left: density per chain; right: draws per chain) of the session-level γ_h, γ_l, β_0 and β_1 of left_bi in sessions 04-06, 04-13 and 05-30.

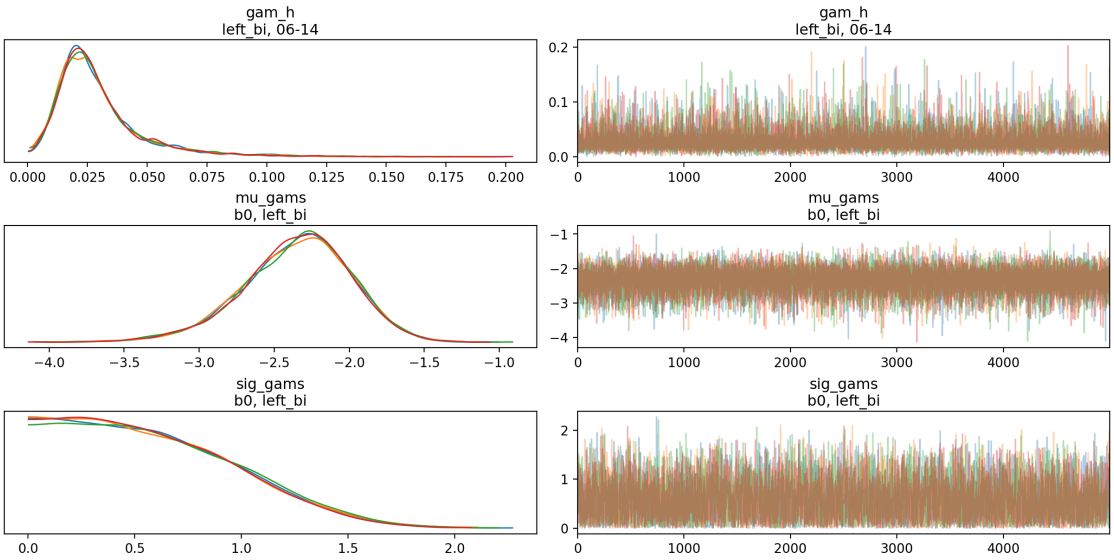


Figure 2: Trace plots of left_bi γ_h on 06-14 and of the hyperparameters μ_{γ_h} (`mu_gams`, b0) and σ_{γ_h} (`sig_gams`, b0) it is drawn from.

2.2 Posterior predictive checks

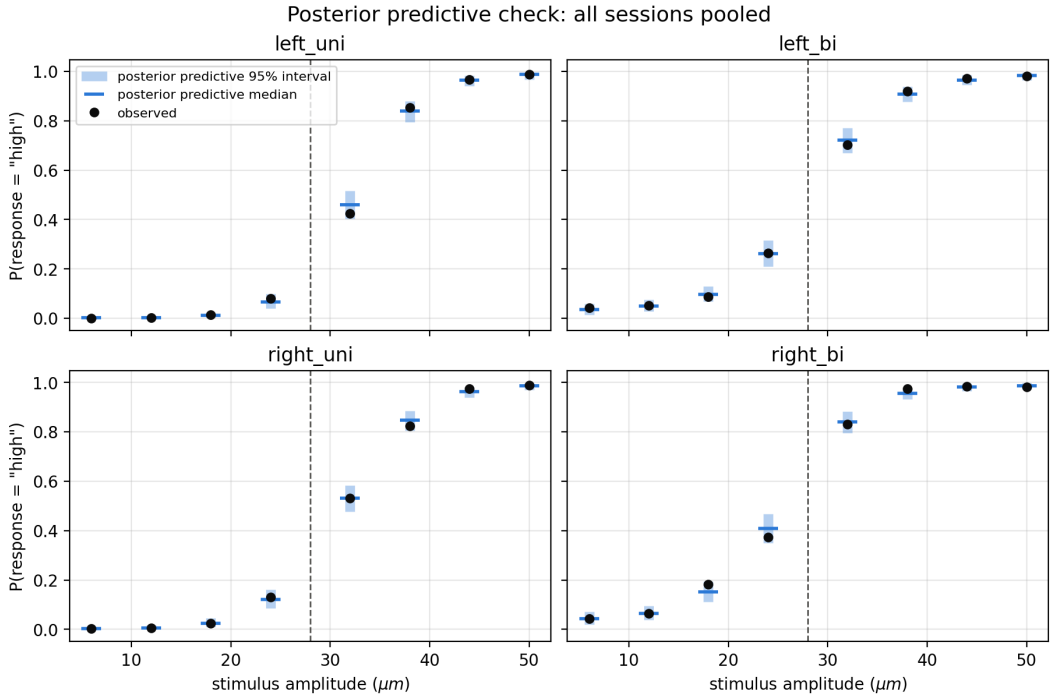


Figure 3: Observed proportion of "high" responses at each amplitude, pooled over sessions (black), against the posterior predictive median and 95% interval (blue). The dashed line is the trained threshold of 28 μm.

All 32 group-by-amplitude proportions lie inside their 95% posterior predictive intervals (Figure 3). At the level of single sessions, 1,341 of the 1,344 session × group × amplitude cells (99.8%) lie inside their intervals. That is more than 95% because each cell's interval comes from a model fit partly to that same cell, so this is a check for gross misfit rather than an exact calibration test; no systematic misfit appears.

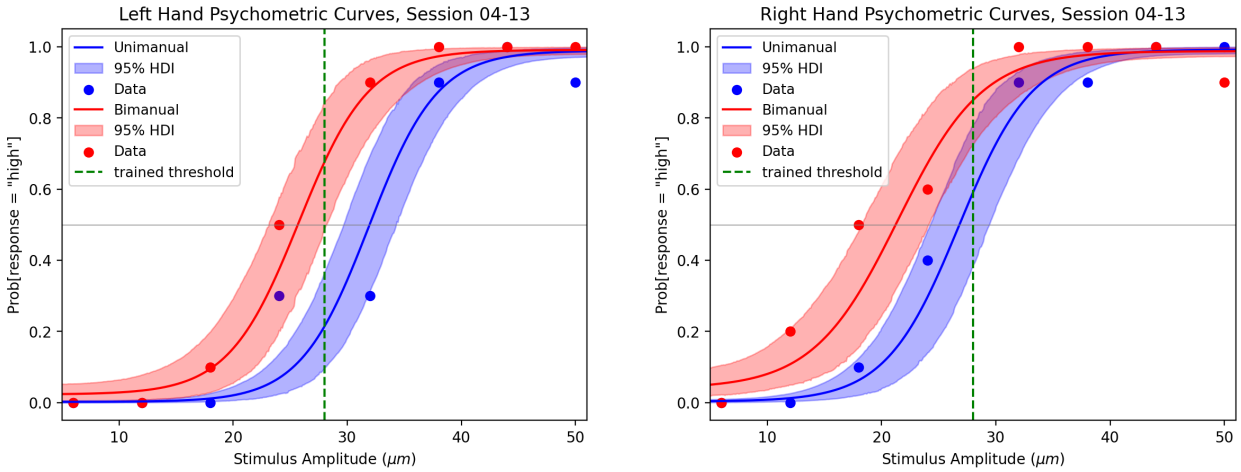


Figure 4: Fitted psychometric curves for session 04-13 (left and right hand): the curve at the posterior means of the parameters, the pointwise 95% HDI of the curve, and the observed proportions of "high" responses (10 trials per point). Unimanual in blue, bimanual in red; the dashed line is the trained threshold.

Figure 4 shows the fit in one session (04-13, the session that `Main_Analysis_B.py` plots). With only 10 trials per point the observed proportions scatter around the curves, but the curves follow them and the bimanual curve lies to the left of the unimanual one for both hands, which is the PSE shift of Section 4. Because every session borrows strength from the others, single stray points (for example the left unimanual proportion of 0.3 at 32 μm) pull the curve only partly towards them.

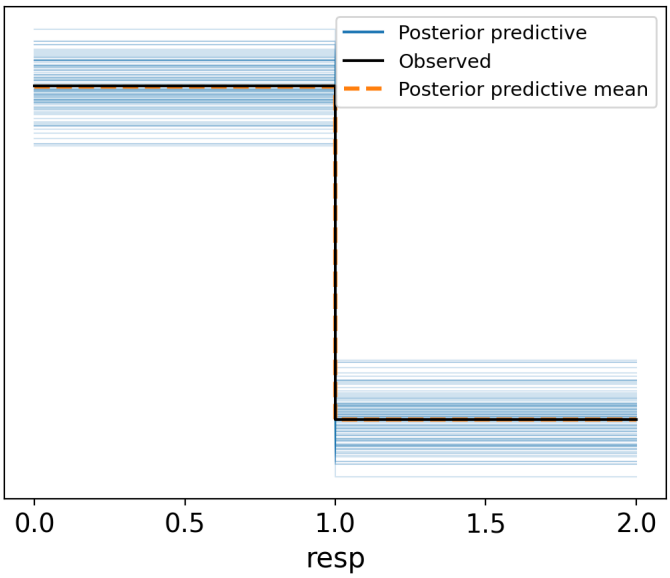


Figure 5: ArviZ posterior predictive check of the responses (0 = "low", 1 = "high"), pooled over all trials: observed (black), 100 posterior predictive draws (blue) and their mean (orange dashed).

The overall proportion of "high" responses is reproduced (Figure 5); for binary data this check is coarse, and Figure 3 is the more informative one.

Leave-one-out cross-validation gives $\mathrm{elpd}_{\mathrm{loo}} = -3180.0$ (SE 64.3) with $p_{\mathrm{loo}} = 179$ and every Pareto k below 0.7. The fit with the earlier priors had -3182.8 (SE 64.2) and $p_{\mathrm{loo}} = 195$: the new priors predict equally well with fewer effective parameters.

3 Population-level Results

Table 1: Typical-session psychometric quantities: the PSE, JND and lapse rates at the group-level means. Posterior mean [95% HDI].

	PSE (μm)	JND (μm)	γ_h	γ_l
left_uni	32.4 [31.6, 33.3]	3.29 [2.95, 3.63]	0.002 [0.000, 0.005]	0.009 [0.000, 0.019]
left_bi	28.2 [27.2, 29.2]	3.89 [3.34, 4.46]	0.023 [0.009, 0.037]	0.009 [0.001, 0.017]
right_uni	31.4 [30.3, 32.5]	3.56 [3.23, 3.89]	0.003 [0.000, 0.006]	0.007 [0.000, 0.015]
right_bi	25.5 [24.5, 26.5]	3.95 [3.48, 4.44]	0.036 [0.017, 0.056]	0.010 [0.003, 0.018]

Without the distractor the PSE lies 3–4 μm above the trained threshold of 28 μm for both hands: the monkey needs a slightly stronger stimulus than the trained boundary before it reports "high". With the distractor the PSE moves to or below the threshold. Lapse rates are small (below 0.04 everywhere); γ_h is close to 0 without the distractor.

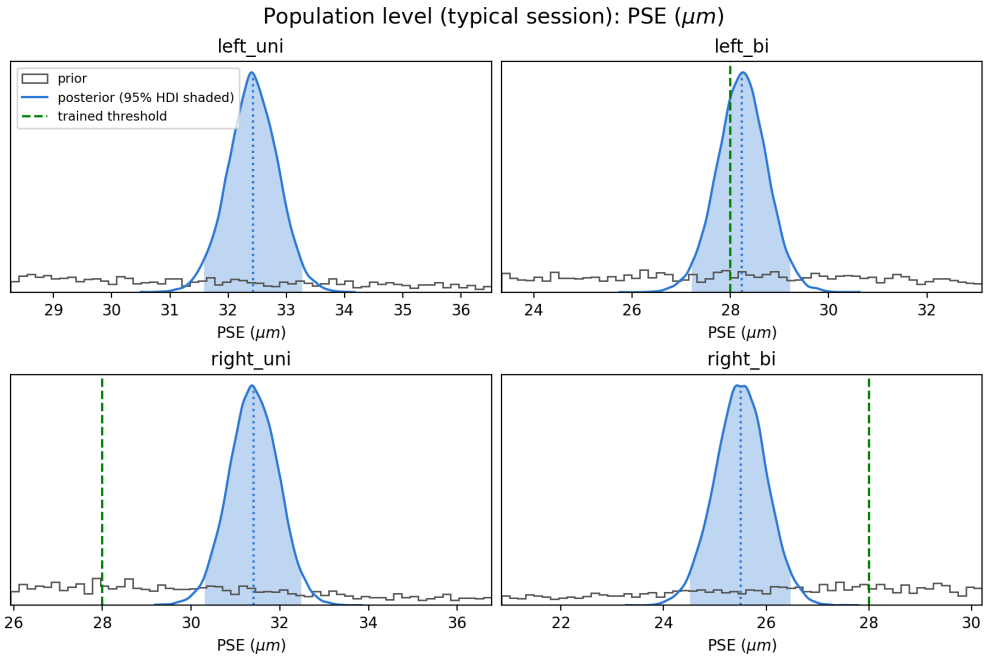


Figure 6: Prior (gray) and posterior (blue) of the typical-session PSE in each group. The posterior is far narrower than the prior, so the data, not the prior, determine it.

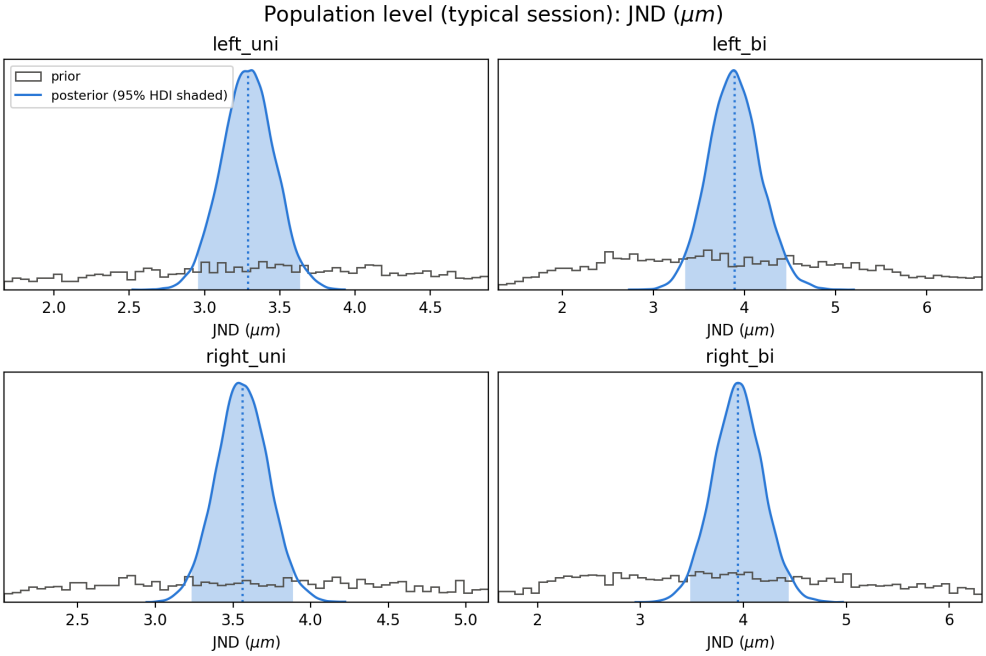


Figure 7: Prior (gray) and posterior (blue) of the typical-session JND.

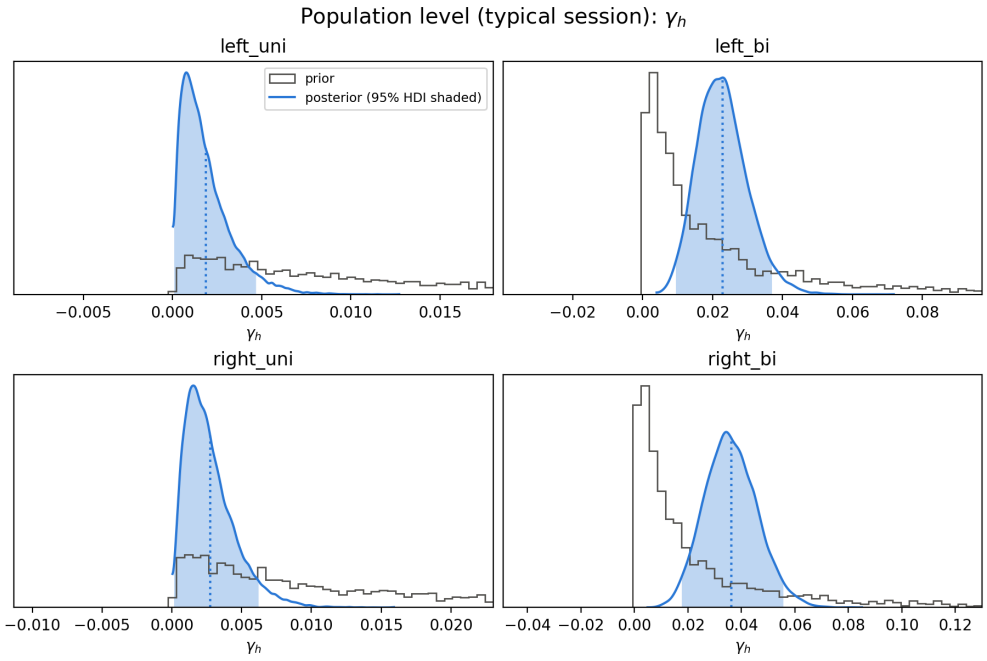


Figure 8: Prior (gray) and posterior (blue) of the typical-session γ_h.

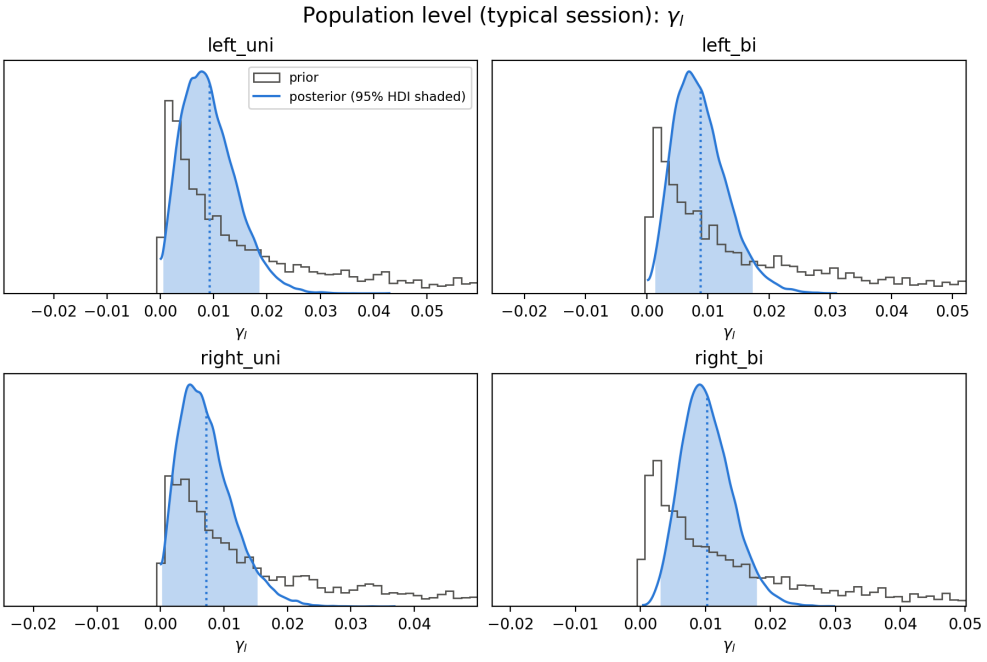


Figure 9: Prior (gray) and posterior (blue) of the typical-session γ_l.

The JND is also determined by the data: its posterior is narrow and the prior is nearly flat over it (Figure 7). The lapse rates are small quantities with priors concentrated near 0, so the comparison is less clear-cut (Figures 8 and 9), but the data still move them. For the bimanual groups the posterior of γ_h lies well to the right of the bulk of the prior, and its 95% HDI excludes

0; for the unimanual groups it is pushed against 0, narrower than the prior.

3.1 Hyperparameters, and comparison with the earlier priors

Table 2: Hyperparameters under the earlier and current priors: posterior mean (sd). Lapse-rate means μ_γ are on the logit scale.

	left_uni		left_bi		right_uni		right_bi	
	earlier	current	earlier	current	earlier	current	earlier	current
μ_{β_1}	5.15 (0.31)	4.97 (0.28)	4.51 (0.34)	4.32 (0.34)	4.69 (0.24)	4.58 (0.23)	4.39 (0.31)	4.34 (0.30)
σ_{β_0}	0.80 (0.16)	0.76 (0.15)	0.77 (0.15)	0.77 (0.14)	1.04 (0.16)	1.01 (0.16)	0.77 (0.15)	0.76 (0.15)
σ_{β_1}	0.32 (0.24)	0.28 (0.22)	1.15 (0.36)	1.27 (0.28)	0.20 (0.15)	0.17 (0.14)	0.66 (0.33)	0.61 (0.32)
μ_{γ_h}	−4.46 (0.57)	−5.20 (0.85)	−2.46 (0.45)	−2.35 (0.36)	−4.26 (0.54)	−4.73 (0.73)	−1.86 (0.36)	−1.82 (0.34)
μ_{γ_l}	−3.21 (0.52)	−3.44 (0.69)	−3.29 (0.50)	−3.44 (0.58)	−3.44 (0.55)	−3.72 (0.73)	−3.26 (0.44)	−3.23 (0.43)
σ_{γ_h}	0.53 (0.41)	0.36 (0.27)	1.54 (0.64)	0.59 (0.40)	0.52 (0.40)	0.37 (0.28)	0.56 (0.40)	0.36 (0.27)
σ_{γ_l}	0.57 (0.43)	0.37 (0.28)	0.74 (0.54)	0.39 (0.29)	0.58 (0.43)	0.38 (0.29)	0.68 (0.45)	0.43 (0.30)

The bias and slope parameters barely change with the new priors (Table 2); the group slopes move down by 0.05–0.19, well within one posterior sd. The group slopes are 4.3–5.0 for every group, below the steep range ($\mu_{\beta_1} \approx 6$) where the convergence study found the slope poorly identified.

The lapse-rate parameters change more. The lapse means μ_{γ_h} of the unimanual groups, which have almost no lapses, move lower (left_uni from -4.46 to -5.20): the new prior $\mathcal{N}(-3, 1.5)$ is wider than the earlier $\mathcal{N}(-2.5, 1)$ and no longer pulls them up. Every between-session lapse sd σ_γ shrinks to 0.36–0.59, close to the new prior (median 0.34). The convergence study showed that σ_{γ_h} and σ_{γ_l} are not identified at this data size, so these values mainly reflect the prior and should not be interpreted. The one notable change is left_bi σ_{γ_h}, which drops from 1.54 to 0.59.

σ_{β_1} for left_bi stands out: 1.27 [0.74, 1.83], far above the other groups, which means the left-hand bimanual slope varies a lot between sessions. A few sessions have very shallow bimanual slopes; the JND of left_bi on 05-12 is 10.9 μm, against a median of 4.1 μm across sessions.

4 Effect of the Distractor

The effect of the distractor is the bimanual − unimanual difference, computed draw by draw so that its uncertainty is exact. It is reported in two ways:

- **average over the 42 sessions:** the mean of the within-session differences, which describes these sessions;
- **population (typical session):** the difference at the group-level means, which describes a new session from the same population and is therefore more uncertain.

Table 3: Distractor effect (bimanual − unimanual). Posterior mean [95% HDI]; the last column counts sessions whose HDI lies entirely above / below 0.

Hand	Quantity	average over sessions	population	sessions > 0 / < 0
Left	PSE (μm)	-4.29 [-5.05, -3.52]	-4.19 [-5.52, -2.90]	0 / 23
	JND (μm)	1.03 [0.46, 1.59]	0.60 [-0.03, 1.28]	5 / 0
	γ_h	0.025 [0.009, 0.042]	0.021 [0.007, 0.035]	7 / 0
	γ_l	0.000 [-0.014, 0.013]	0.000 [-0.014, 0.012]	0 / 0
Right	PSE (μm)	-6.06 [-6.80, -5.34]	-5.91 [-7.34, -4.40]	0 / 32
	JND (μm)	0.49 [-0.07, 1.03]	0.39 [-0.19, 0.96]	0 / 0
	γ_h	0.035 [0.016, 0.054]	0.033 [0.014, 0.053]	42 / 0
	γ_l	0.003 [-0.009, 0.015]	0.003 [-0.009, 0.014]	0 / 0

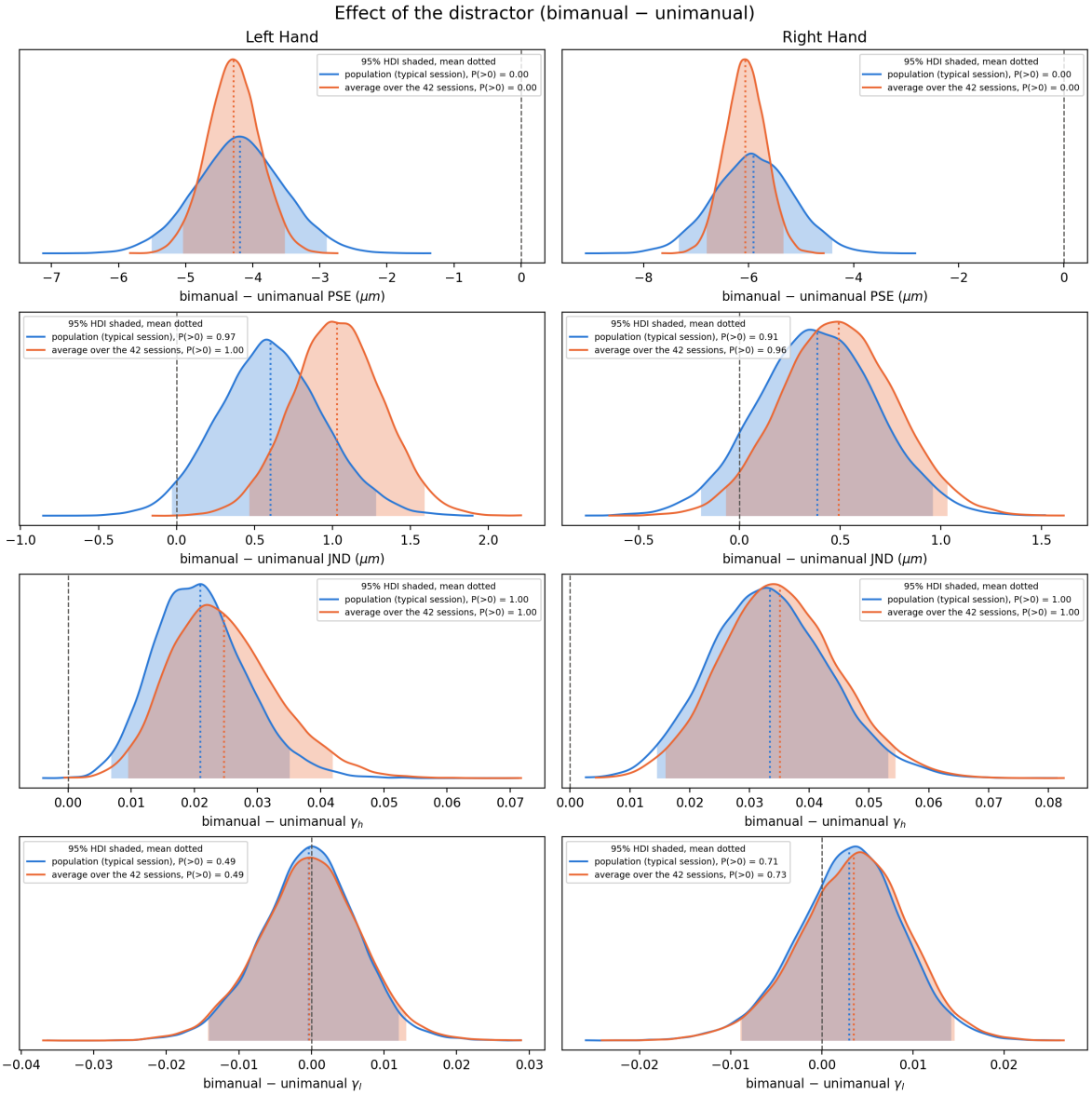


Figure 10: Posterior of the distractor effect: population level (blue) and average over the 42 sessions (orange), with 95% HDIs shaded.

PSE. The clearest effect. The distractor lowers the PSE by about 4 μm for the left hand and 6 μm for the right hand (Table 3). The posterior mean of the shift is negative in every right-hand session and in 40 of 42 left-hand sessions; the two exceptions are early, weak-distractor sessions with shifts below +0.5 μm (Figure 11). This is consistent with a stimulus feeling stronger when a distractor is applied to the other hand.

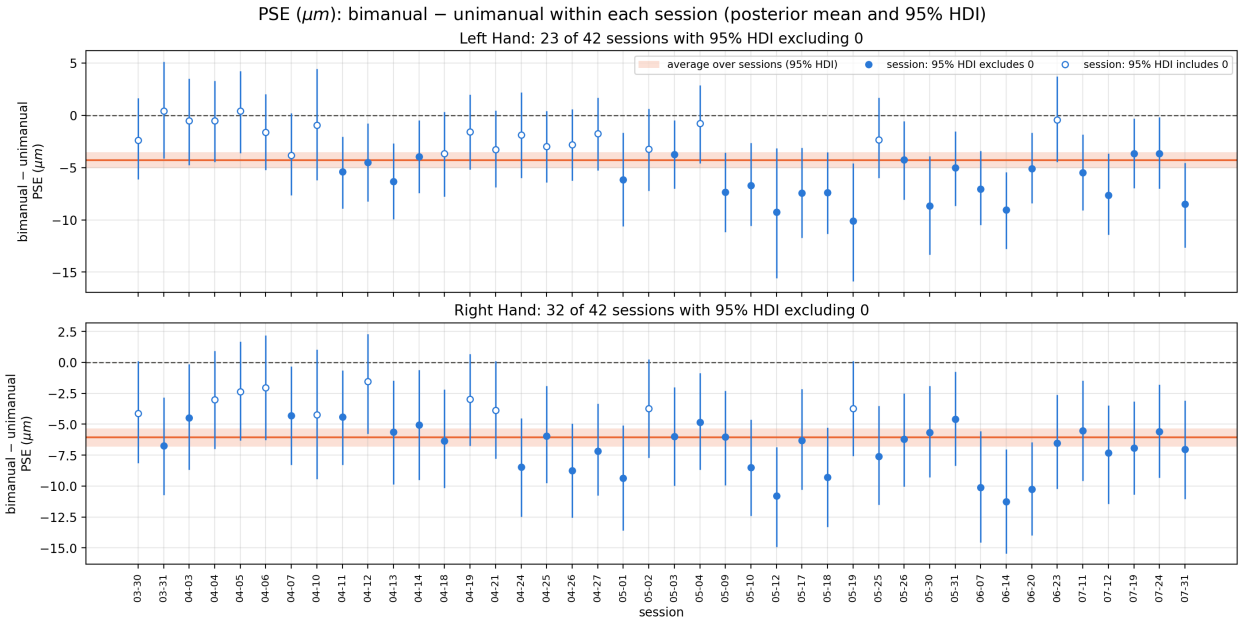


Figure 11: PSE shift (bimanual − unimanual) in each session: posterior mean and 95% HDI. Filled points: HDI excludes 0. The orange band is the average over sessions.

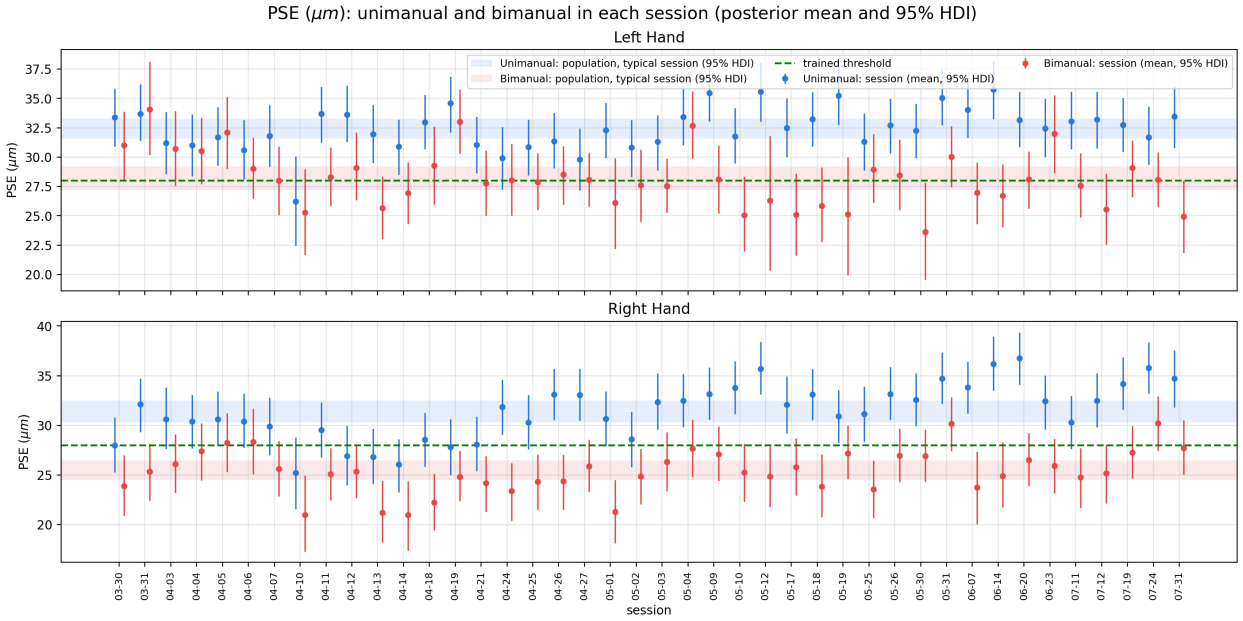


Figure 12: Unimanual (blue) and bimanual (red) PSE in each session, with the population-level typical-session HDIs as bands and the trained threshold (green).

Figure 12 shows the two conditions separately. The unimanual PSE of the right hand rises over the study, from about 27 μm in early April to about 35 μm in June and July, while the bimanual PSE stays between 21 and 30 μm. This is part of the amplitude dependence discussed in Section 4.1.

γ_h. With the distractor the monkey more often reports "high" for the weakest stimuli: γ_h rises from about 0.002 to 0.02–0.04 (Table 1). The rise is clear in every right-hand session (Figure 13). This agrees with the PSE shift: the distractor adds to the perceived strength of the stimulus, enough to make even the weakest stimuli occasionally seem "high". The convergence study found left_bi γ_h estimated too low in some datasets, so the left-hand effect may be somewhat larger than reported here, not smaller.

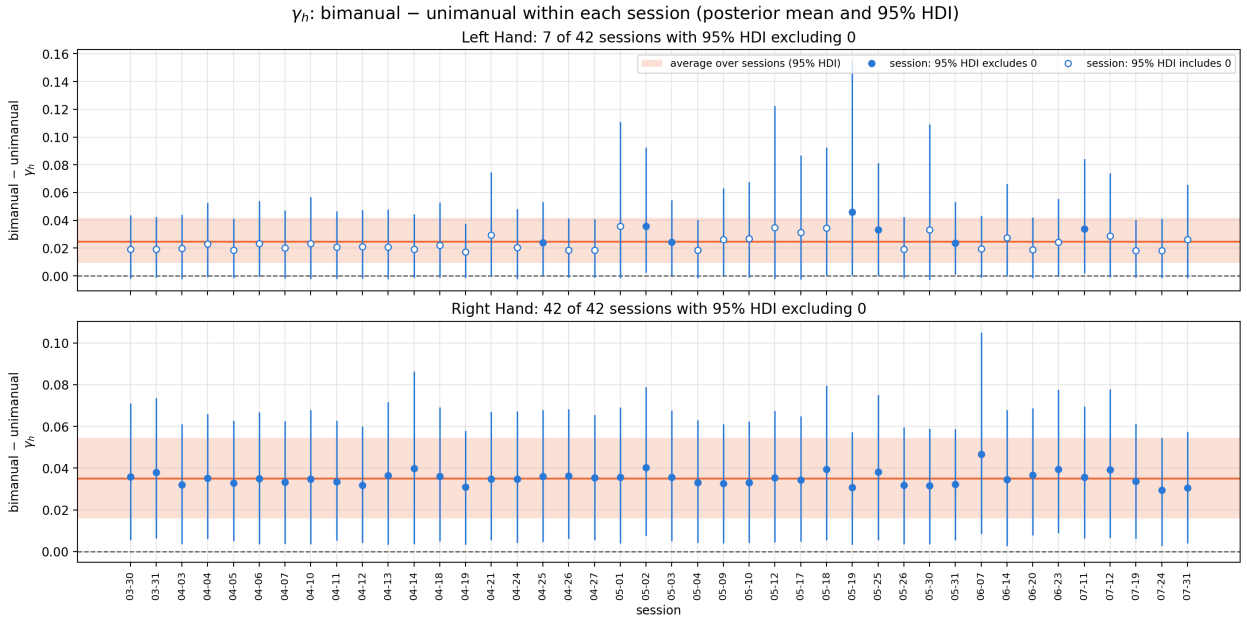


Figure 13: γ_h shift (bimanual − unimanual) in each session.

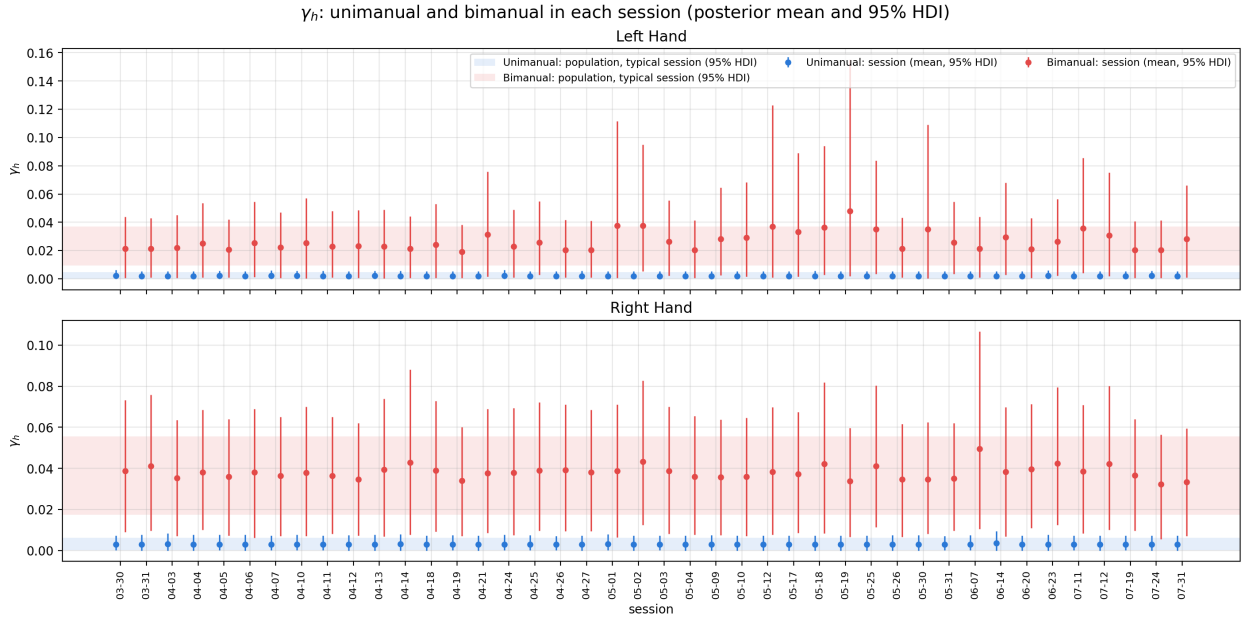


Figure 14: Unimanual (blue) and bimanual (red) γ_h in each session, with the population-level typical-session HDIs as bands.

The session estimates of γ_h (Figure 14) are pulled strongly towards their group's typical-session value: the bimanual posterior means range only from 0.019 to 0.048 (left) and 0.032 to 0.050 (right). Each session has few trials at the weakest amplitudes, and the between-session sd σ_{γ_h} is small and mostly set by its prior, so session-to-session differences in γ_h cannot be resolved; the clear result is the difference between conditions, not differences between sessions.

JND. The distractor makes discrimination slightly less precise. For the left hand the average over sessions is $+1.0$ μm [0.5, 1.6], but this is driven by a handful of sessions with very shallow bimanual curves (Figure 15, e.g. 05-12 at $+7.6$ μm), consistent with the large σ_{β_1} of left_bi. At the population level the effect is $+0.6$ μm [-0.03, 1.28], with $P(\text{effect} > 0) = 0.97$. For the right hand the effect is about $+0.4$–0.5 μm with HDIs that include 0, and no single session's HDI excludes 0.

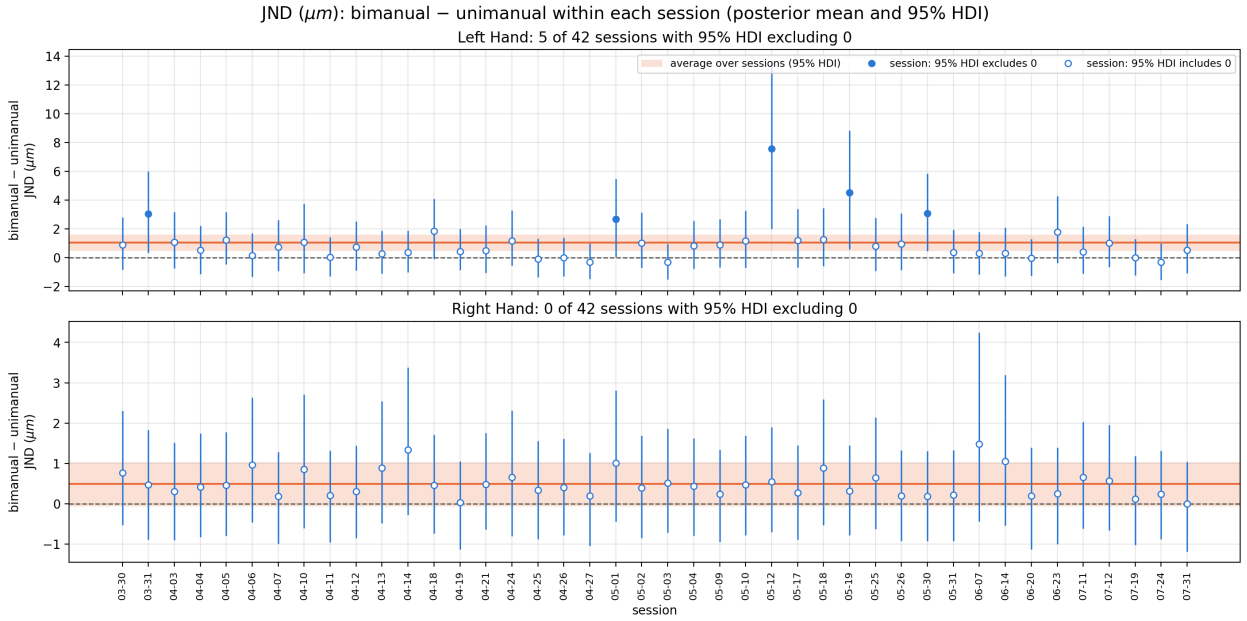


Figure 15: JND shift (bimanual − unimanual) in each session.

γ_l**.** No effect for either hand (posterior means 0.000 and 0.003, HDIs centered near 0; Figures 16 and 17). Like γ_h, the session estimates of γ_l barely vary between sessions (posterior means 0.007–0.017).

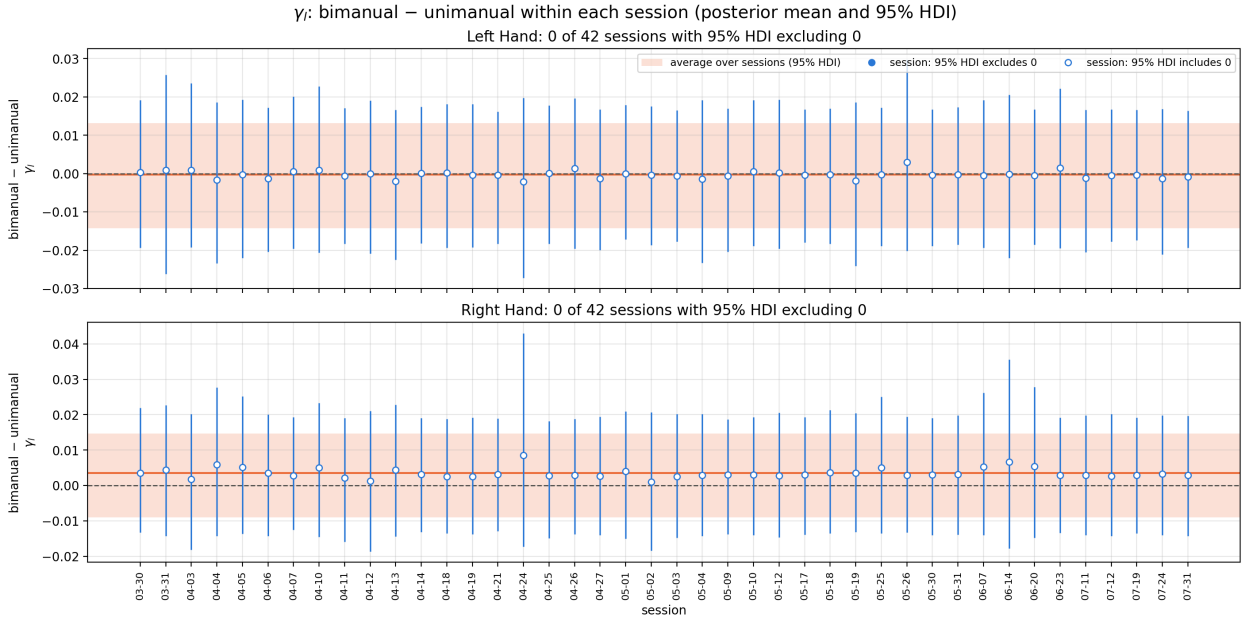


Figure 16: γ_l shift (bimanual − unimanual) in each session.

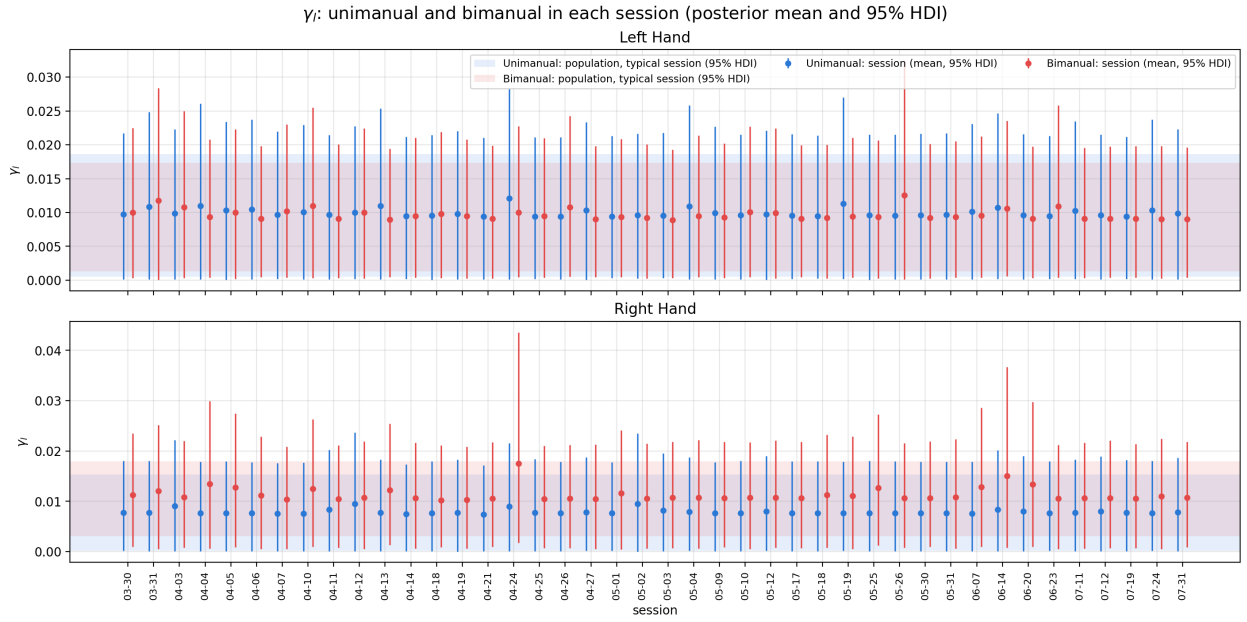


Figure 17: Unimanual (blue) and bimanual (red) γ_l in each session, with the population-level typical-session HDIs as bands.

4.1 Dependence on distractor amplitude (exploratory)

The distractor amplitude was not constant: it was 6 μm in the first 8 sessions, 12 μm in the next 6, increased gradually to 32 μm over the next 11 (three of them at 24 μm), and then stayed at 32 μm with occasional sessions at 24 μm: 13 sessions at 32 and 7 at 24 in total. The model does not include distractor amplitude, so this section looks at the fitted session-level effects afterwards, with

a linear regression of the session shifts on amplitude computed separately for each posterior draw.

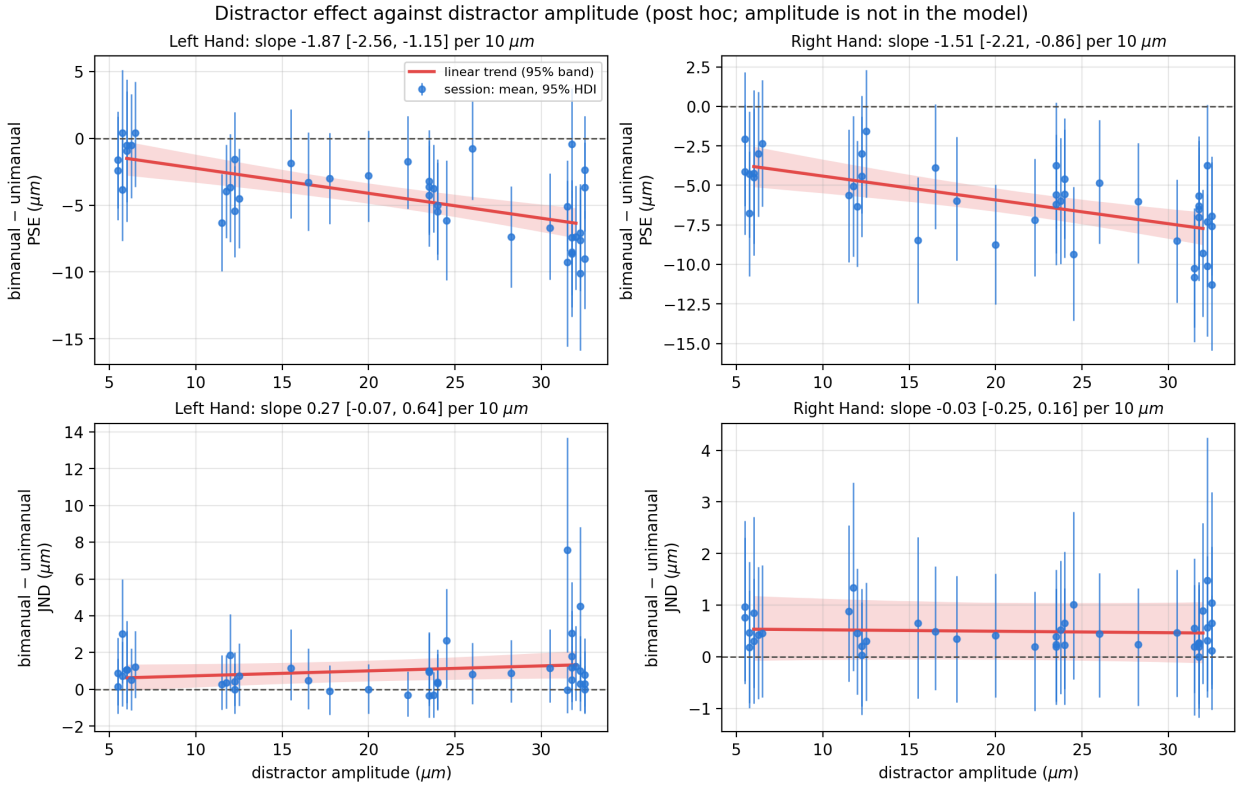


Figure 18: Session-level PSE and JND shifts against distractor amplitude (points slightly spread horizontally where several sessions share an amplitude), with the per-draw linear trend and its 95% band.

Table 4: Mean PSE shift (bimanual − unimanual, μm) over the sessions at each distractor amplitude.

Distractor amplitude	6 μm (8 sessions)	12 μm (6)	24 μm (7)	32 μm (13)
Left hand	−1.1 [−2.8, 0.4]	−4.2 [−5.7, −2.7]	−4.5 [−6.0, −3.0]	−6.7 [−8.0, −5.3]
Right hand	−3.9 [−5.5, −2.4]	−4.3 [−6.1, −2.6]	−5.9 [−7.4, −4.3]	−7.9 [−9.1, −6.7]

The PSE shift grows with distractor amplitude, by −1.9 μm [−2.6, −1.2] (left) and −1.5 μm [−2.2, −0.9] (right) per 10 μm of distractor (Figure 18, Table 4). At the weakest distractor the left-hand shift is not distinguishable from 0. The JND shift shows no clear trend with amplitude.

Because amplitude rose over the study, amplitude and session order are strongly correlated ($r = 0.91$), so a change over time (learning, adaptation) could produce the same pattern. The later alternation between 24 and 32 μm partly separates the two. A regression on both amplitude and session order attributes the trend to amplitude (−2.4 μm [−4.0, −0.8] per 10 μm for the left hand, −2.1 [−3.6, −0.7] for the right) and finds no trend with session order (+0.5 [−0.7, 1.6] and +0.6 [−0.6, 1.8] μm per 10 sessions). Split by condition, the bimanual PSEs fall with amplitude and the unimanual PSEs rise somewhat with it, so a stronger distractor also seems to shift the unimanual trials of the same session (for example through the monkey's criterion adapting to the session as

a whole). Both right-hand PSEs also drift upward with session order (unimanual +0.8 [0.0, 1.6], bimanual +1.4 [0.5, 2.3] μm per 10 sessions), a drift that largely cancels in the difference. With 42 sessions and two strongly correlated predictors these conclusions are tentative; a model with distractor amplitude as a session-level covariate would test them properly.

5 Conclusions

- The fit is reliable: no divergences, $\hat{R} = 1.00$ and large ESS for every quantity of interest, every Pareto k good, and no systematic misfit in the posterior predictive checks.
- The distractor shifts the PSE down by 4–6 μm and raises γ_h by 0.02–0.035, both clearly and consistently across sessions, consistent with a distractor on the other hand making the stimulus feel stronger.
- The distractor slightly worsens discrimination (JND up by about 0.5–1 μm), most clearly for the left hand, where it is driven by a few sessions.
- The PSE shift appears to grow with distractor amplitude, by about 2 μm per 10 μm; this needs confirming with amplitude in the model.
- The new priors change the bias and slope estimates very little. The between-session lapse sds now sit near their prior, which is expected because the convergence study showed that they are not identified.

Possible follow-ups.

1. Add distractor amplitude as a session-level covariate on the bimanual β_0 (and possibly on the unimanual one), to test the amplitude dependence inside the model.
2. Define σ_{β_0} as a HalfNormal variable directly to remove the $\hat{R}$ warning on `z_sig_b0`.
3. Look at the left_bi sessions with very shallow slopes (05-12, 05-19, 05-30) for problems with the data or the session.

A Joint Posteriors

`Main_Analysis_B.py` also plots joint posteriors for session 06-14 (Figures 19–23). The correlation of β_0 and β_1 follows from the stimulus axis being centered on 28 μm: it is negative ($r \approx -0.4$) for the unimanual groups, whose PSE on this day lies above 28 μm (36 μm), and positive ($r = 0.1$–0.3) for the bimanual groups, whose PSE lies below it (25–27 μm). The lapse rates are only weakly correlated with β_0 and β_1 ($|r| \leq 0.26$, mostly with the slope) and with each other. There is no ridge or multimodality that would indicate an identifiability problem.

Figure 23 looks at the left_bi γ_h together with the hyperparameters it is drawn from. γ_h is positively correlated with both μ_{γ_h} ($r = 0.34$) and σ_{γ_h} ($r = 0.29$): a larger session value is easier to reach when the group mean is higher or the spread around it is wider. σ_{γ_h} has its mode at 0 and is barely constrained by the data, as the convergence study found. The sampler had no trouble with this region (no divergences).

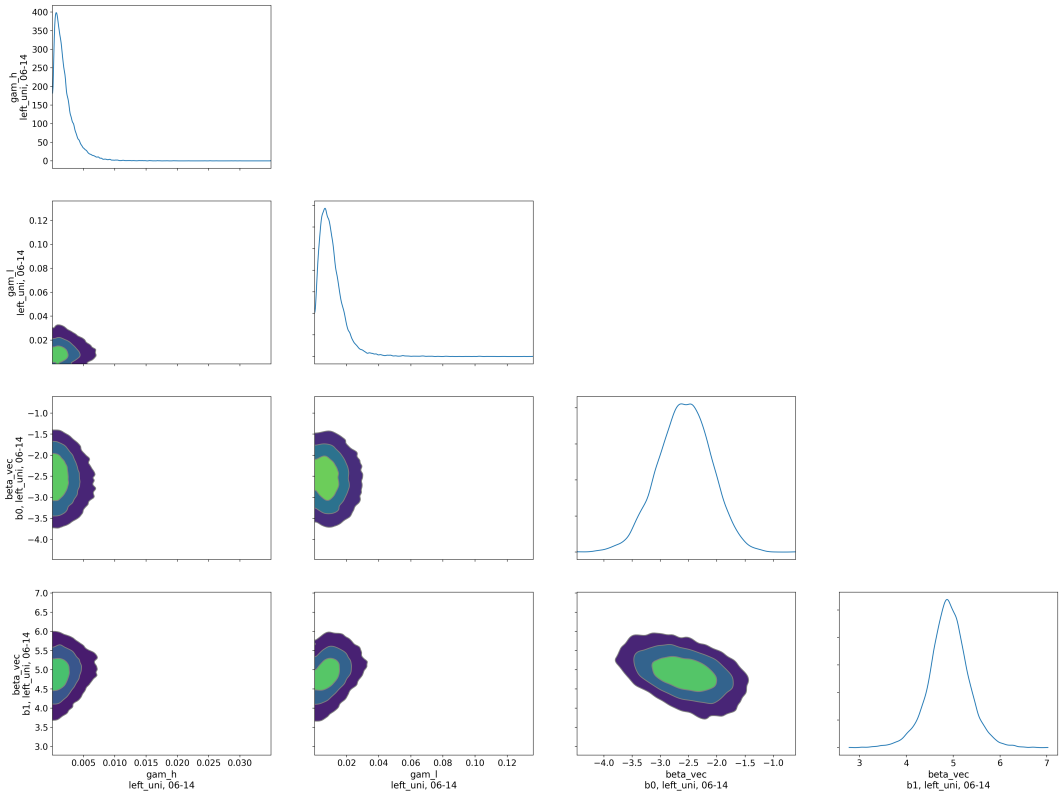


Figure 19: Joint posterior of γ_h, γ_l, β_0 and β_1 for left_uni, session 06-14.

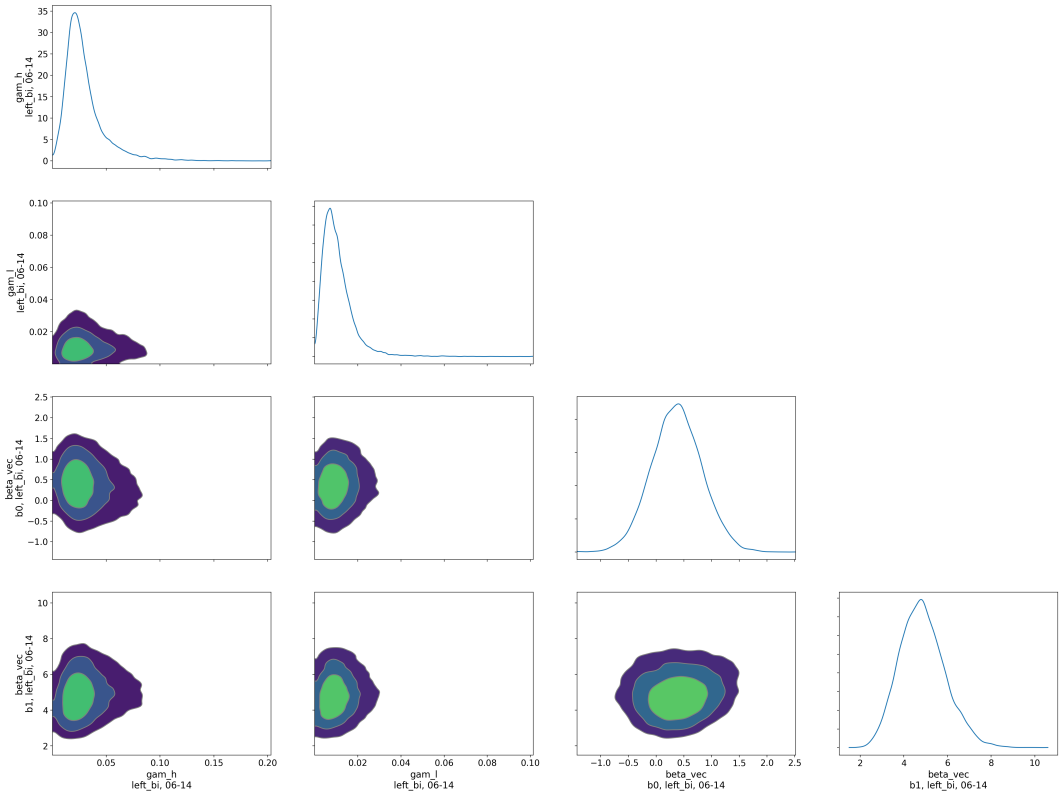


Figure 20: As Figure 19, for left_bi.

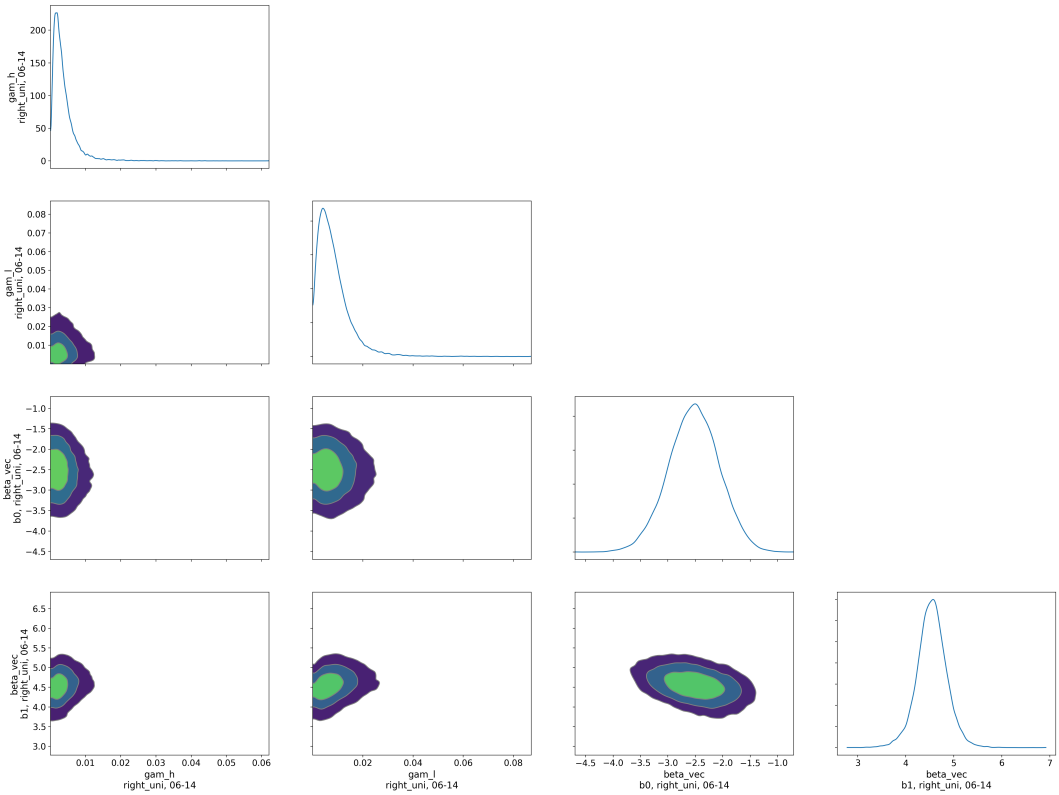


Figure 21: As Figure 19, for right_uni.

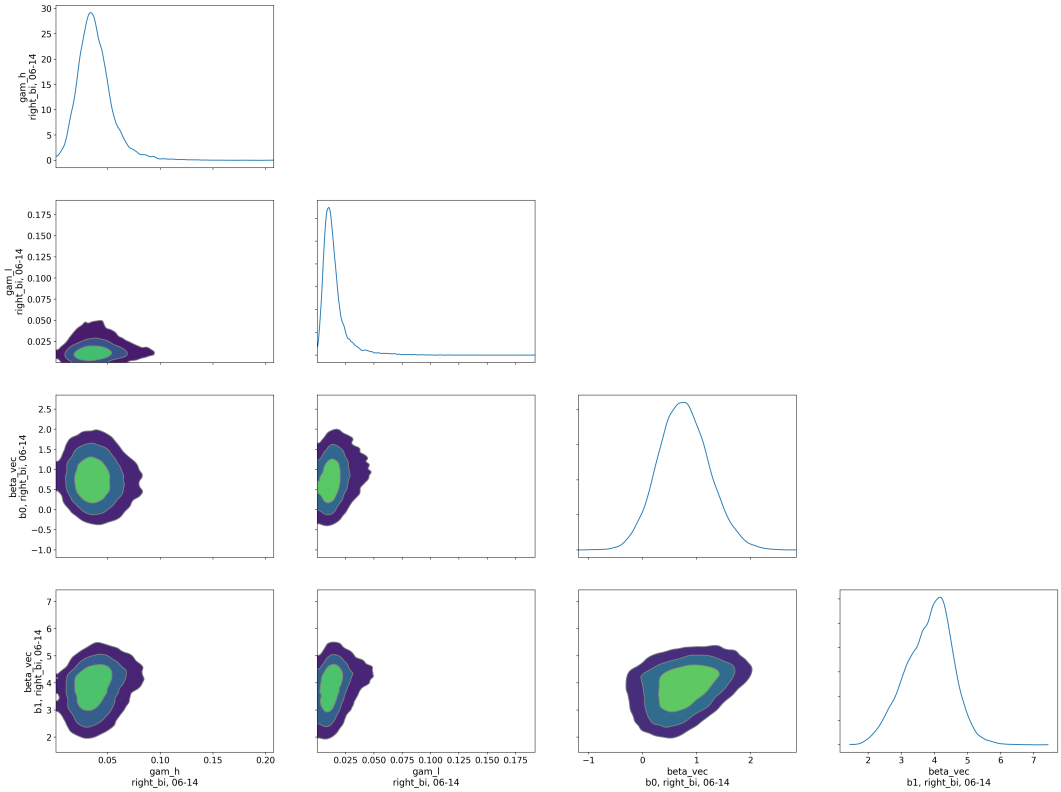


Figure 22: As Figure 19, for right_bi.

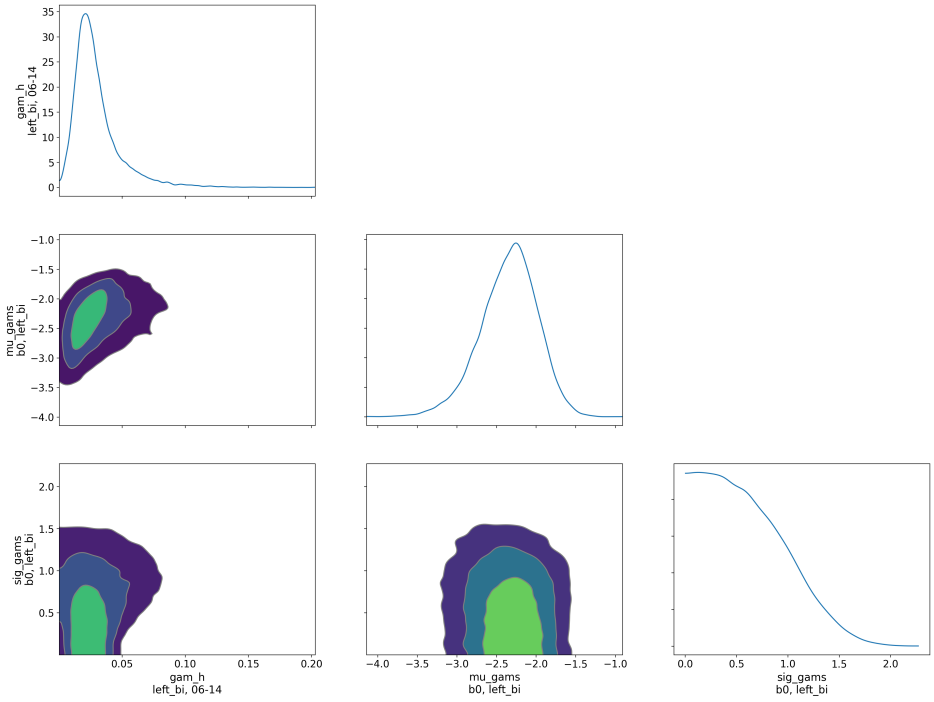


Figure 23: Joint posterior of left_bi γ_h on 06-14 and the hyperparameters μ_{γ_h} and σ_{γ_h}.